Assignment 2.1

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Q1) (a) Radiation force in a 1D MOT is given by

$$\vec{F} = \hbar\vec{k}_1\left[R_{sc}\left(\delta + \vec{k}_1\cdot\vec{v} + \frac{\mu\vec{B}\cdot\hat{z}}{\hbar}\right) - R_{sc}\left(\delta - \vec{k}_1\cdot\vec{v} - \frac{\mu\vec{B}\cdot\hat{z}}{\hbar}\right)\right]$$

Here,

σ^+ $\vec{k}_1$ $\vec{v}$ $\vec{k}_2$ σ^- $\qquad \vec{k}_1 = -\vec{k}_2 = k\hat{x}$

Now, assuming 1D, $\vec{k}_1\cdot\vec{v} = kv$

Now, for $F=0 \rightarrow F'=1$ transition, $\mu = (g_{F'}m_{F'} - g_F m_F)\mu_B$

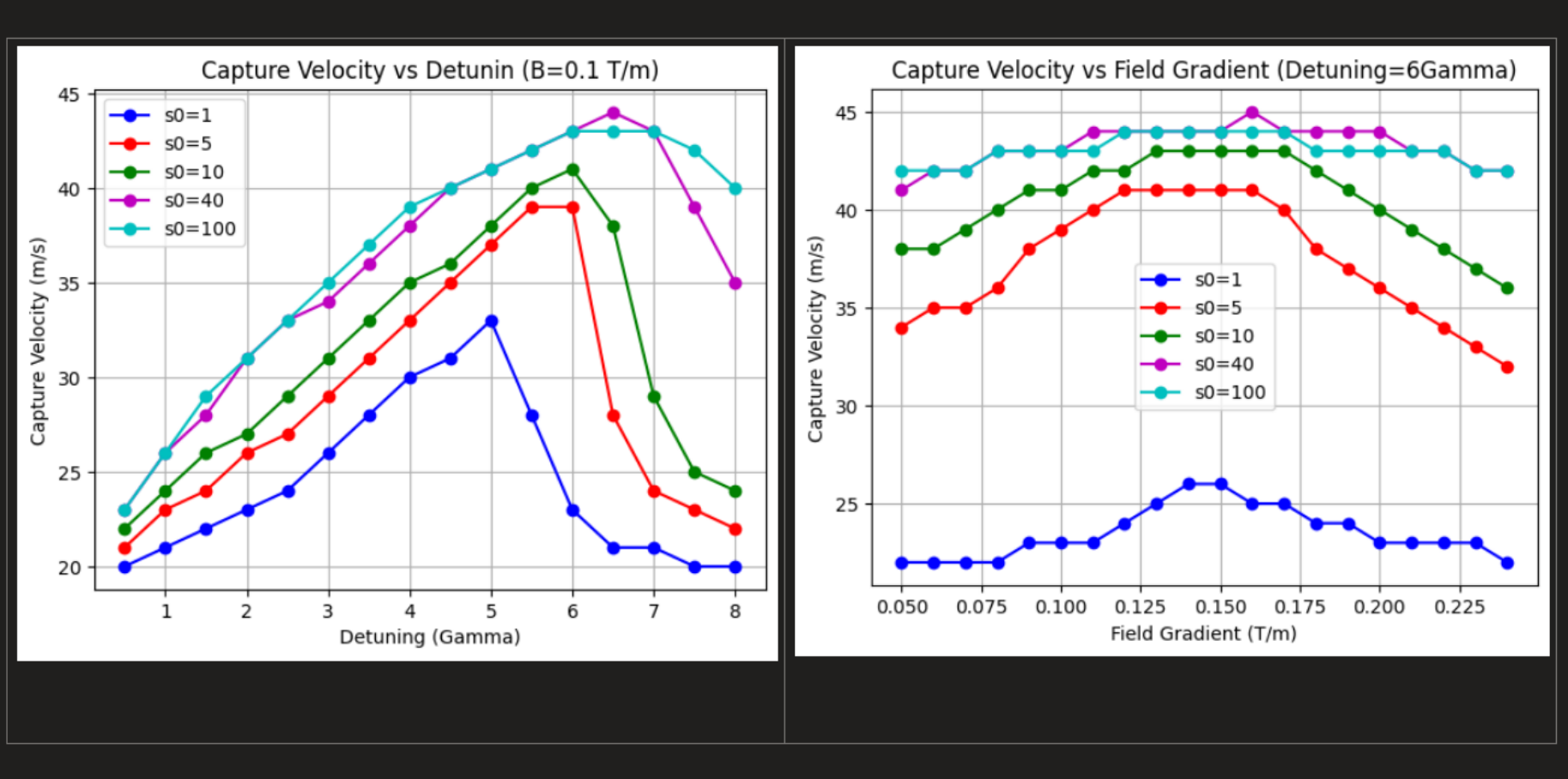
$\Rightarrow m_F = 0$, $m_F' = -1, 0, 1$ but σ^+ & σ^- beams pick ± 1. $g_{F'=1} = 1$

$\Rightarrow \mu = \mu_B$

$$\Rightarrow m\ddot{z} = \hbar k\left[R_{sc}\overbrace{\left(\delta + kv + \frac{\mu B' z}{\hbar}\right)}^{\Delta_1} - R_{sc}\overbrace{\left(\delta - kv - \frac{\mu B' z}{\hbar}\right)}^{\Delta_2}\right]$$

$$R_{sc}(\Delta) = \frac{\Gamma}{2}\frac{s_0}{1 + s_0 + 4\Delta^2/\Gamma^2} \quad \text{and} \quad s = \frac{I}{I_{sat}}$$

B' is the field gradient. All calculations considered with Cs atoms D2 transition



$\Rightarrow$ For 1D, the capture velocity is in the range of 40 - 45 m/s

(b) acceleration is given by $a = \frac{\hbar k}{m}\cdot\frac{\Gamma}{2} \approx 5.85\times10^4\ m/s^2$

$$\text{Length} = \frac{v_f^2 - v_i^2}{2a} = \frac{(200)^2 - (30)^2}{1.17\times10^5} \approx \boxed{0.334\, m} \approx 34\ cm$$

(c) Each spontaneous decay event imparts a momentum of $\hbar k$ in a random direction.
To calculate the number of such events, we note that the recoil velocity in axial direction

is $v_r = \frac{\hbar k}{m} = 3.5\times10^{-3}\ m/s$

$\Rightarrow$ To go from 200 m/s to 30 m/s you must have

$$\frac{200-30}{3.5\times10^{-3}} = 4.82\times10^4.$$

According to random walk, $\langle p^2\rangle = N(\hbar k)^2$

And in the transverse direction, $\langle p_\perp^2\rangle = \frac{2N}{3}(\hbar k)^2$

$$\Rightarrow \langle v_\perp\rangle = \sqrt{\frac{2N}{3}}\cdot\frac{\hbar k}{m} = 0.63\ m/s$$

$$\Rightarrow k_B T_\perp = 2\times\frac{1}{2}m\langle v_\perp^2\rangle$$

$$\Rightarrow \boxed{T_\perp = 6.4\ mK}$$

Q2) $\alpha_0 = 1151$ a.u. , $\lambda = 1064$ nm

a) trap depth = 100 μK

From slides, $U_{FORT} = \frac{\alpha^{(0)}(\omega)}{2c\varepsilon_0}I = \frac{\alpha^{(0)}_{a.u.}\times 4\pi\cancel{\varepsilon_0}a_0^3}{2c\cancel{\varepsilon_0}}\cdot I$

$$\Rightarrow I_{max} = \frac{U_{FORT,\mu K}\times k_B\times 2c}{4\pi a_0^3\times\alpha^{(0)}_{a.u.}} = \boxed{3.86\ KW/cm^2}$$

(b)

$$U_{FORT}(r,0) = \frac{\alpha^{(0)}}{2c\varepsilon_0}\cdot\frac{2P}{\pi w^2}e^{-2r^2/w^2} \underset{r\to0}{\approx} \frac{\alpha^{(0)}}{2c\varepsilon_0}\cdot\frac{2P}{\pi w^2}\cdot\left(1 - \frac{2r^2}{w^2}\right) = U_0\left(1 - \frac{2r^2}{w^2}\right)$$

$\|^{ly}$

$$U_{FORT}(0,z) = \frac{\alpha^{(0)}}{2c\varepsilon_0}\cdot\frac{2P}{\pi w^2}\cdot\frac{1}{1+\left(\frac{z}{z_R}\right)^2} \underset{z\to0}{\approx} \frac{\alpha^{(0)}}{2c\varepsilon_0}\cdot\frac{2P}{\pi w^2}\left(1 - \frac{z^2}{z_R^2}\right) = U_0\left(1 - \frac{z^2}{z_R^2}\right)$$

$$z_R = \frac{\pi w^2}{\lambda}$$

where $U_0 = \frac{\alpha^{(0)}\cdot 2P}{2c\varepsilon_0\cdot\pi w^2} = 100\,\mu k\cdot k_B$

Thus trap frequencies are

$$\omega_r = \sqrt{\frac{4U_0}{mw^2}} \qquad \omega_z = \sqrt{\frac{2U_0}{mz_R^2}}$$

$$\boxed{\begin{aligned}\omega_r &= 2\pi\times0.5\ KHz\\ \omega_z &= 2\pi\times2.4\ Hz\end{aligned}}$$

(c) Assuming beam is along gravity (z- direction)

$$\frac{dU}{dz} = U_0\cdot\frac{2z/z_R^2}{\left(1+(z/z_R)^2\right)^2}$$

This is maximum when $\frac{d^2U}{dz^2} = 0 \Rightarrow \frac{1}{\left(1+(z/z_R)^2\right)^2} = \frac{4z^2/z_R^2}{\left(1+(z/z_R)^2\right)^3}$

$$\Rightarrow \boxed{z_{max} = \frac{z_R}{\sqrt{3}}}$$

$$\Rightarrow F_{max} = -U_0\cdot\frac{2}{\sqrt{3}z_R}\cdot\frac{9}{16} = \frac{-3\sqrt{3}}{8}\cdot\frac{U_0}{z_R}$$

$$= \boxed{1.215\times10^{-25}\ N}$$

$F_{gravity} = mg = 2.16\times10^{-24}$ N

$\Rightarrow$ it will not be sufficient if the beam is vertical

If the beam is horizontal, $F_{max} = \left|\frac{dU}{dr}\right|_{max} = \left|U_0\cdot\left(\frac{-4r}{w_0^2}\right)\cdot e^{-2r^2/w_0^2}\right|_{max}$

$\Rightarrow e^{-2r^2/w^2} - \frac{4r^2}{w^2}e^{-2r^2/w^2} = 0 \Rightarrow r_{max} = w/2 \Rightarrow F_{max} = \frac{2U_0}{w}\cdot e^{-1/2} = \boxed{3.35\times10^{-23}\ N}$

$\Rightarrow$ it will hold if beam is horizontal.

Q3) $E = \int_0^\infty \varepsilon\, n(\varepsilon)\rho(\varepsilon)\, d\varepsilon$, $V(r) = \frac{1}{2}\omega^2 r^2$

$n(\varepsilon) = \frac{1}{e^{(\varepsilon-\mu)/k_BT} - 1}$ (Bose Einstein distribution

$\rho(\varepsilon) = \frac{dG(\varepsilon)}{d\varepsilon} = \frac{\varepsilon^2}{2(\hbar\omega)^3}$ (for 3D harmonic trap)

$$\Rightarrow E = \int_0^\infty \frac{\varepsilon}{e^{(\varepsilon-\mu)/k_BT} - 1}\cdot\frac{\varepsilon^2}{2(\hbar\omega)^3}\, d\varepsilon$$

$$= \frac{1}{2(\hbar\omega)^3}\cdot\int_0^\infty \frac{\varepsilon^3}{e^{(\varepsilon-\mu)/k_BT} - 1}\, d\varepsilon$$

$$= \frac{(k_BT)^4}{2(\hbar\omega)^3}\int_0^\infty \frac{x^{4-1}}{z^{-1}e^x - 1}\, dx \quad \text{where } z = e^{\mu/k_BT}$$

$$\boxed{E = \frac{(k_BT)^4}{2(\hbar\omega)^3}\cdot\Gamma(4)\,Li_4(z)} = 3k_BT\cdot\left(\frac{k_BT}{\hbar\omega}\right)^3 Li_4(z)$$

Now, for 3D $\boxed{N_{th} = \frac{(k_BT)^3}{2(\hbar\omega)^3}\cdot\Gamma(3)\,Li_3(z)}$

For 3D harmonic trap, critical temperature is given when $N_{tot} = N_{th}$

$$N_{tot} = \frac{(k_BT_c)^3}{\cancel{2}(\hbar\omega)^3}\cancel{\Gamma(3)}\zeta(3) \qquad \zeta(3) = N_{tot}\frac{(\hbar\omega)^3}{(k_BT_c)^3}$$

$$\Rightarrow (\hbar\omega)^3 = \frac{\zeta(3)(k_BT_c)^3}{N_{tot}}$$

$$\frac{N_{tot}t^3}{\zeta(3)} = \left(\frac{k_BT}{\hbar\omega}\right)^3$$

$t = T/T_c$ (used in code)

Although $N_{tot} = N_0 + N_{th}$

and $N_0 = n(0) = \frac{1}{z^{-1}e^0 - 1} = \frac{z}{1-z}$

$$\Rightarrow \boxed{N_{tot} = \frac{z}{1-z} + \left(\frac{k_BT}{\hbar\omega}\right)^3 Li_3(z)}$$

Note: $\frac{d\,Li_\alpha(z)}{dT} = \frac{d}{dT}\sum_1^\infty z^n n^{-\alpha}$

$= \sum_1^\infty n z^{n-1}\cdot n^{-\alpha}\cdot\frac{dz}{dT}$

$= \frac{1}{z}\left(\sum_1^\infty z^n n^{-\alpha+1}\right)\cdot\frac{dz}{dT}$

$= \frac{Li_{\alpha-1}(z)}{z}\cdot\frac{dz}{dT}$

Now, $\frac{dN}{dT} = \frac{1}{(1-z)^2}\cdot\frac{dz}{dT} + \left(\frac{k_B}{\hbar\omega}\right)^3\left(3T^2Li_3(z) + T^3\cdot\frac{1}{z}Li_2(z)\cdot\frac{dz}{dT}\right)$

$= 0$

as N must be conserved

$$\Rightarrow \boxed{\left[\frac{1}{(1-z)^2} + \left(\frac{k_BT}{\hbar\omega}\right)^3\frac{Li_2(z)}{z}\right]\frac{dz}{dT} = \frac{-3}{T}\left(\frac{k_BT}{\hbar\omega}\right)^3 Li_3(z)}$$

Now, $dE/dT = 3k_B\left(\frac{k_B}{\hbar\omega}\right)^3\left[4T^3Li_4(z) + \frac{T^4}{z}Li_3(z)\frac{dz}{dT}\right]$

We know $N_{th} = \left(\frac{k_BT}{\hbar\omega}\right)^3 Li_3(z)$

$$\Rightarrow C = 3k_B\left[4N_{th}\cdot\frac{Li_4(z)}{Li_3(z)} + \frac{T}{z}\cdot N_{th}\cdot\frac{-3/T\cdot N_{Th}}{\frac{1}{(1-z)^2} + \frac{N_{th}}{z}\cdot\frac{Li_2(z)}{Li_3(z)}}\right]$$

$$\boxed{C = k_B\left[12N_{th}\frac{Li_4(z)}{Li_3(z)} - \frac{9N_{th}^2}{N_0(1+N_0) + N_{th}\frac{Li_2(z)}{Li_3(z)}}\right]}$$

$$\left(\frac{z}{(1-z)^2} = N_0\cdot\frac{1}{1-z} = N_0\left(\frac{z}{1-z}+1\right) = N_0(N_0+1)\right)$$

(i) for $T < T_c$, $z = 1$

$$\Rightarrow C = 12k_BN_{th}\cdot\frac{\zeta(4)}{\zeta(3)} - \frac{O(\cancel{N_{th}})}{\cancel{O(N_0^2)}}\cdot N_{th}$$

(as at $T < T_c$, most atoms occupy ground state and $N_0 \sim N_{tot} \Rightarrow N_0 \gg N_{th}$)

Thus $\boxed{C_{T<T_c} = 12k_BN_{th}\frac{\zeta(4)}{\zeta(3)}}$ $\qquad$ Also $\frac{N_{tot}}{N_{th}} = \left(\frac{T_c}{T}\right)^3$

$$= 12k_B\cdot N_{tot}\left(\frac{T}{T_c}\right)^3\cdot\frac{\zeta(4)}{\zeta(3)}$$

$$\Rightarrow \boxed{C_{T<T_c} = 10.8\,N_{tot}k_B\left(\frac{T}{T_c}\right)^3}$$

(ii) for $T > T_c$, $N_{th} \gg N_0$

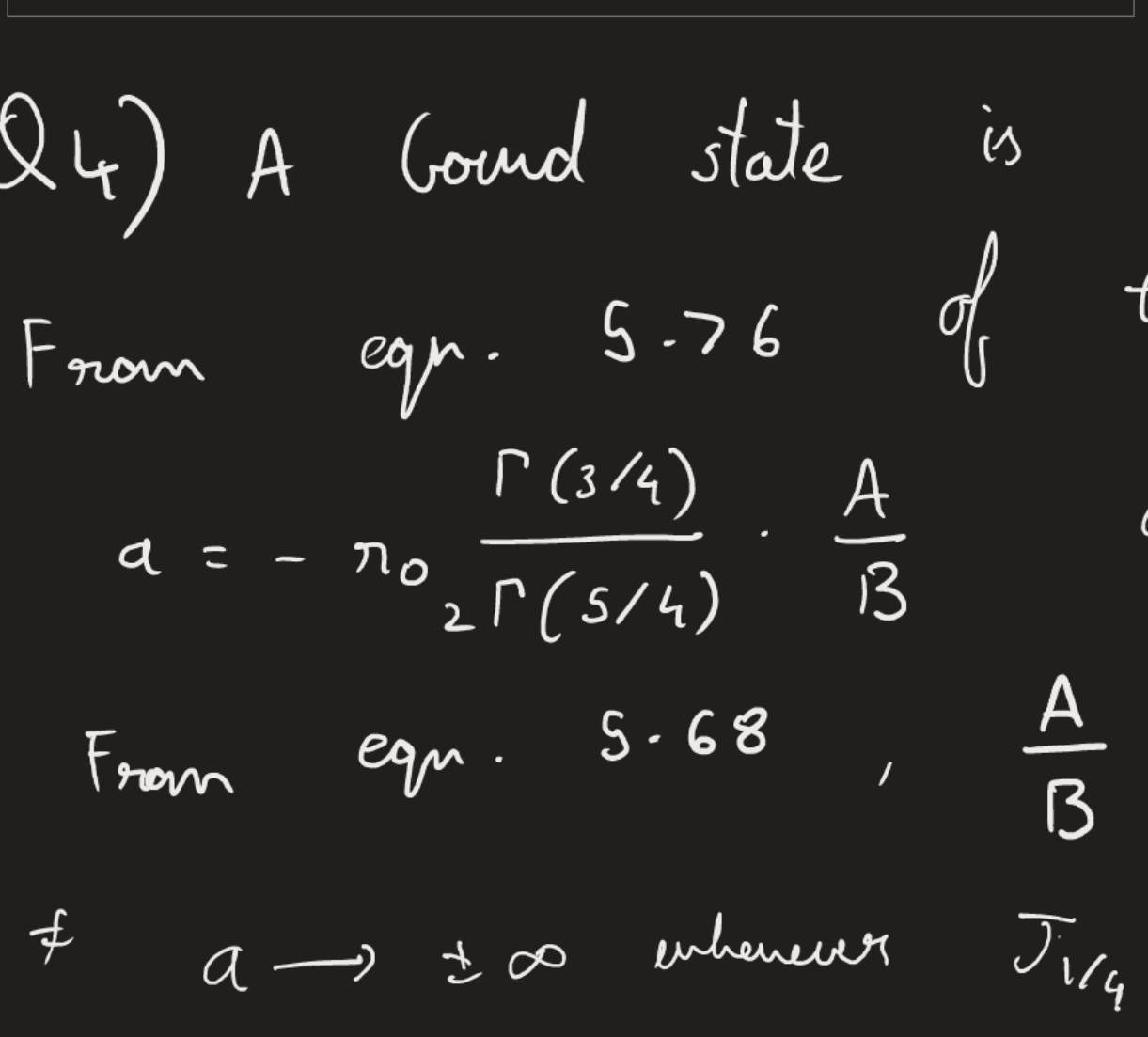
$$\Rightarrow C = 12k_BN_{th}\frac{Li_4(z)}{Li_3(z)} - \frac{9N_{th}^2k_B}{N_{th}\cdot\frac{Li_2(z)}{Li_3(z)}} \qquad \text{here } N_{tot} \approx N_{th}$$

$$\boxed{C_{T>T_c} = k_BN_{tot}\left[12\frac{Li_4(z)}{Li_3(z)} - \frac{9\,Li_3(z)}{Li_2(z)}\right]} \quad \text{where } Li_3(z) = \zeta(3)\left(\frac{T_c}{T}\right)^3$$

$\Rightarrow$ as $T\to\infty$, $z\to0 \Rightarrow Li_n(z)\to z \Rightarrow C\to 3Nk_B$ as expected according to equipartition

as $T\to T_c^+$, $z\to1 \Rightarrow C\to[10.8 - 6.58)Nk_B = 4.23\,Nk_B$

$\Rightarrow$ there is a sharp drop of C at T_c.



Q4) A bound state is formed whenever scattering length $a \rightarrow \pm\infty$

From eqn. 5.76 of textbook,

$$a = -r_0\frac{\Gamma(3/4)}{2\Gamma(5/4)}\cdot\frac{A}{B} \quad \text{where } r_0 = \left(\frac{2M_r\alpha}{\hbar^2}\right)^{1/4}$$

From eqn. 5.68, $\frac{A}{B} = \frac{-J_{-1/4}(\phi)}{J_{1/4}(\phi)}$ where $\phi = \frac{r_0^2}{2r_c^2}$ & J's are Bessel functions

$\Rightarrow a\rightarrow\pm\infty$ whenever $J_{1/4}(\phi)\rightarrow0$

By asymptotic expansion methods, Zeroes of Bessel functions can be calculated exactly. First zero occurs at $\phi \approx 2.7809$

$$\Rightarrow \frac{r_c}{r_0} = \frac{1}{\sqrt{2\phi}} \Rightarrow \boxed{\frac{r_c}{r_0} = 0.424}$$

for large z approximation also we arrive at almost the same value

$\Rightarrow \cos\left(\phi - \frac{3\pi}{8}\right) = 0 \Rightarrow \phi \approx \frac{7\pi}{8} \approx 2.75 \Rightarrow \frac{r_c}{r_0} \approx 0.426$

This means if $\overset{\text{core radius } r_c}{\wedge} > 0.424\,r_0$, there will be no bound state. Every subsequent bound state would be at larger ϕ i.e. smaller r_c.

(ii) From equation 5.1, length scale r_0 is given by

$$r_0 \sim (C_6 m/m_e)^{1/4}a_0$$

From table 5.1., C_6 of H-H is 6.5 a.u.

$\Rightarrow r_0 \approx 10.45a_0$

$\Rightarrow$ for bound state, $r_c < 0.424\times r_0 = 4.43\,a_0$

Although, it is given that the potential is repulsive $< 7a_0$ (the potential is positive). This means $r_c > 7a_0$ which is $> 4.43a_0$

Thus, $\phi = \frac{r_0^2}{2r_c^2} \approx \frac{10.5^2}{2\times7^2} \approx 1.1 < 2.78$

Thus, it is impossible to form a bound state for 2 spin-polarized H atoms in triplet state.
This happens because Hydrogen is very light and has small polarizability (small m and $C_6 \Rightarrow$ small r_0).

Code: https://github.com/ReyshaJV/Ultracold-atoms/blob/main/assignment2.ipynb